

9 (a) By varying the resistance of the variable resistor R from its maximum resistance to zero **B1**

(b) (i) False, because the resistances shown on the graph are not constant. **B1**

OR

To test Ohm's law, the same resistor/resistance must be used.

OR

To test Ohm's law, the current through a fixed resistance is measured as the p.d. across it is varied.

(ii) False. Battery does not supply electrons, but an electric field / electromotive force / potential difference / electrical energy to cause the free electrons already in the circuit to move. **B1**

(c) (i) The graph is a plot of $V = E - Ir$.
Hence, the V-intercept is $E = 3.00 \text{ V}$. **A1**

(ii) Gradient $= -r = \frac{3.00 - 0.20}{0.0 - 4.0} = \frac{2.80}{-4.0} = -0.70 \Omega$ **C1**

So $r = 0.70 \Omega$ **A1**

Deduct 1 mark if gradient is positive or if r is negative.

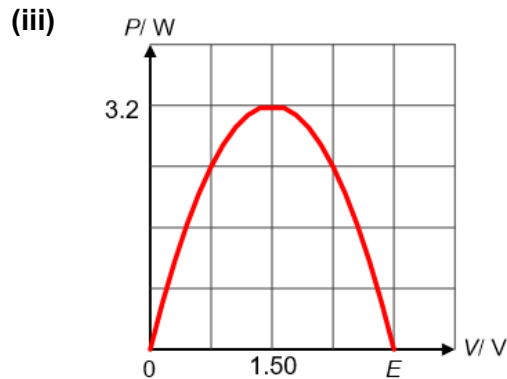
- (d) (i) Power output is maximum when $R = r = 0.70 \Omega$. A1
Using potential divider rule, $V = 1.5 \text{ V}$ (allow ECF)

(ii)
$$P_{\max} = \frac{V^2}{R} = \frac{1.5^2}{0.70} \quad (\text{allow ECF}) \quad \text{M1}$$
$$= 3.2 \text{ W} \quad \text{A1}$$

Alternatively:

Draw a line of gradient 0.70Ω to determine values of V and I at point of intersection $(2.1, 1.50)$ when R is 0.70Ω . M1

Hence, $P_{\max} = 2.1 \text{ A} \times 1.50 \text{ V} = 3.2 \text{ W}$ (to 2 s.f.) A1



Quadratic graph opens down. B1
Symmetrical about $V = E/2$ (peak is at 1.50 V , 3.2 W). B1

(e) (i)
$$\eta = \frac{IV}{IE} = \frac{V}{E} = \frac{1.50}{3.00} = 50\% \quad \text{A1}$$

- (ii) Maximum efficiency is achieved when value of R is maximum, i.e., 10Ω . (Reject $R = \infty$.)

Proof:

$$\eta = \frac{I^2 R}{IE} = I \frac{R}{E} = \left(\frac{E}{r+R} \right) \frac{R}{E} = \frac{R}{r+R} = \frac{1}{r/R + 1} \quad \text{A1}$$

Hence, for efficiency η to approach 100%, the term r/R in the denominator must approach zero, that is, R must be large (which would mean lower power output) or r must be small (not a marking point here).

- (iii) As the purpose of a sound system is to produce as the loudest sound, one would want a maximum power transfer. **B2**

High efficiency is incorrect because it produces a low power output (and of course low power lost, which is irrelevant to this question).

Note: Both points must be correct. Any contradiction in answers will score 0.

- (f) (i) Using potential divider rule, $V_{PQ} = \frac{6.00}{6.00 + 4.00} \times 2.00 = 1.20 \text{ V}$ **M1**

$$E' = \frac{0.600}{1.000} \times 1.20 = 0.720 \text{ V} \quad \textbf{A1}$$

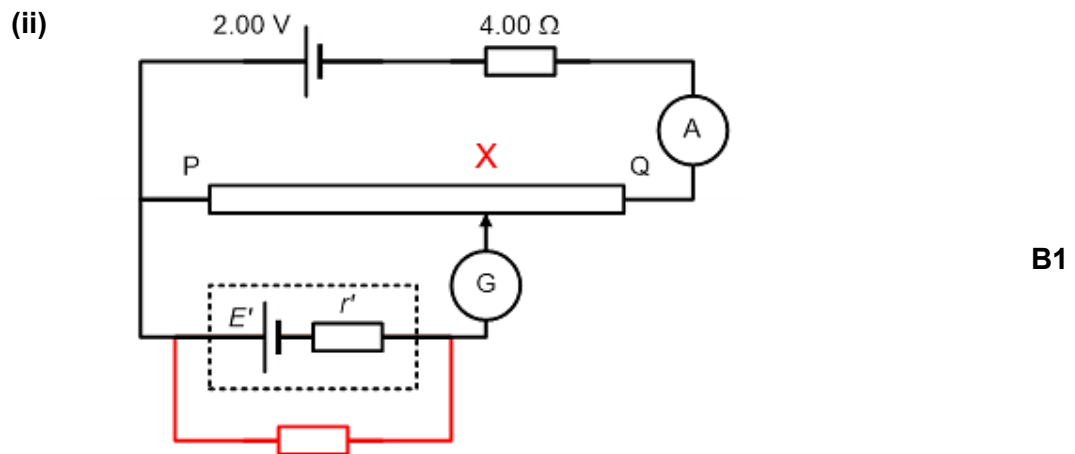


Fig. 9.4

Connect a resistor across the lower branch to allow current to flow through r' so that its value can be determined from $V_{PQ} = E' + Ir'$.

- (iii) From $Q = It$,

$$2000 \times 10^{-3} \text{ Ah} = (0.080 \text{ A}) t$$

$$\Rightarrow t = 25 \text{ h} \quad \textbf{C1}$$

A1